

Graded Project : Part 3 - Explainable AI

⇒Goal

The goal of this part 3 is to implement XAI (Explainable AI) on your project done in part 2

⇒Tools

It is recommended to use a linux or mac OS, an IDE with a **recent python version**.

⇒References

Add all sources you may rely on in your report

⇒Restitution

- Use the same github/gitlab repository as for part 2
- Add a section about your work in the same report as for part 2
- Report must be available in the **same group dedicated repository** :
 AI PM_Graded Project_Groups_042026
- You must share your github code repository with email : dinamedy@hotmail.com

Implement Explainable AI (XAI) for your project with SHAP

Integrate **SHAP Library** to your Project :

- Install SHAP in your python environment (don't forget to add it to your lib requirements)
- Use it to explain your model predictions :
 - Build a **TreeExplainer** and compute Shapley Values
 - Visualize explanations for a specific point of your data set,
 - Visualize explanations for all points of your data set at once,
 - Visualize a **summary plot** for each class on the whole dataset.
 - Go further and generate for a specific data set point :
 - Waterfall plot
 - Force plots
 - Mean SHAP plot
 - Beeswarm plot
 - Dependence plots

The evaluation of this part 3 will be mainly based on the description of your experiments in the report + your project repository on github or gitlab containing :

- The code of XAI (with SHAP)
- XAI outputs